International Journal of Information Security (2025) 24:119
https://doi.org/10.1007/s10207-025-01032-0

REGULAR CONTRIBUTION

Human factors in cybersecurity: an interdisciplinary review and framework proposal

Kalam Khadka[1] · **Abu Barkat Ullah**[1]

Published online: 29 April 2025
© The Author(s) 2025

Abstract
Cybersecurity has traditionally been viewed as a technological challenge, but growing evidence emphasizes the pivotal role of human factors in shaping both vulnerabilities and resilience. This survey explores the critical intersection of human behaviour, decision-making, and organizational culture in cybersecurity, highlighting the need for strategies that effectively integrate technical and human-centric approaches. Key findings reveal persistent gaps, such as the underrepresentation of underserved communities, limited exploration of non-digital social engineering, and unresolved ethical challenges in Artificial Intelligence (AI)-driven cybersecurity tools. Additionally, existing frameworks often lack a standardized, implementable framework for assessing cybersecurity culture and fail to address generational and cultural diversity. These gaps underline the urgency of adaptive, inclusive solutions that align human behaviour with technical systems. To address these challenges, the proposed Human-Centric Cybersecurity Framework integrates psychological resilience, adaptive training, socio-technical approach, and ethical AI principles. The framework outlines practical strategies, including gamified learning, emotional intelligence training, and decision-support systems, to enhance cybersecurity awareness, reduce vulnerabilities, and promote organizational compliance. This paper advocates for an interdisciplinary approach to cybersecurity, emphasizing the importance of empowering individuals and fostering a culture of security. By bridging the gap between human and technical factors, it offers actionable insights for researchers and practitioners to build resilient and inclusive cybersecurity ecosystems capable of countering evolving threats.

Keywords Human factors in cybersecurity · Cybersecurity culture · Socio-technical system · Adaptive training and education · Ethical AI in cybersecurity · Organisational behaviour

1 Introduction

The role of human factors in cybersecurity has gained increasing recognition as technological advancements alone prove insufficient to address vulnerabilities stemming from human behaviour. While cybersecurity is often viewed as a predominantly technical field, even human interactions, decision-making processes, and organizational culture significantly influence digital security systems' effectiveness and resilience. Understanding these human elements is no longer complementary but essential for developing robust, adaptive cybersecurity strategies.

Human behaviour has emerged as a major vulnerability, frequently exploited in sophisticated cyberattacks like social engineering and phishing [1]. High-profile breaches, such as the 2020 Twitter hack, demonstrate how attackers can bypass advanced technological safeguards by targeting human vulnerabilities [2]. These incidents emphasize the necessity of adopting a human-centric approach that integrates psychological, organizational, and behavioural insights into cybersecurity practices [3]. Even the most advanced technical defences are prone to failure without addressing the human component.

Research increasingly focuses on key human factors contributing to cybersecurity risks, such as user awareness, trust, privacy perceptions, and cognitive biases. Studies show that human error whether from negligence, overconfidence, or insufficient training accounts for a significant proportion

✉ Kalam Khadka
kalam.khadka@canberra.edu.au

Abu Barkat Ullah
abu.barkatullah@canberra.edu.au

[1] Faculty of Science and Technology, University of Canberra, Canberra, Australia

of security breaches [4]. Addressing these vulnerabilities requires a shift from technology driven solutions toward strategies that balance both human and technical defences.

The evolving nature of the digital landscape further complicates cybersecurity challenges. Remote work, sophisticated attack vectors, and a rapidly changing threat environment demand innovative approaches that consider the human element in every aspect of security design. Researchers advocate for systems that reduce cognitive strain and enhance user decision-making through interactive training tools, adaptive authentication mechanisms, and behaviour-aligned security systems [5]. These solutions aim to bridge the gap between organizational strategies and technical defences, positioning human factors as a critical component of cybersecurity resilience [6].

While interest in human-centric cybersecurity is increasing, critical gaps persist in understanding the interplay between psychological and organizational factors within technical systems. To bridge these gaps, this study explores key vulnerabilities, identifies underexplored areas, and proposes an integrated framework that enhances cybersecurity resilience. Emerging interdisciplinary research in fields such as behavioural science, criminology, and organizational psychology offers valuable insights for designing targeted interventions, simulations, and training programs that mitigate human-driven cyber risks [7].

The objectives of this paper are threefold:

- To synthesize existing research on human factors in cybersecurity, providing a clear and thorough overview of advancements across disciplines such as psychology, education, organizational studies, and technology. This synthesis helps illustrate the critical role of human behaviour in both strengthening and compromising cybersecurity efforts.
- To identify key gaps in the current literature, including overlooked areas like non-digital social engineering, challenges in ethical Artificial Intelligence (AI) integration, and the lack of focus on underserved populations. These gaps highlight the need for further exploration and targeted solutions in areas that have significant real-world implications.
- To propose a Human-Centric Cybersecurity Framework that directly addresses these gaps. By integrating insights from multiple fields, this framework aims to provide practical strategies for organizations to enhance cybersecurity resilience. It emphasizes inclusivity, adaptability, and real-world application to meet the challenges of diverse organizational and cultural environments.

By pursuing these objectives, this paper contributes to the growing body of research on the importance of human factors in cybersecurity, highlighting actionable strategies for building more resilient systems. The proposed framework aims to harmonize human behaviour, technology, and organizational processes, shifting the focus from blaming users to empowering them as key defenders in the cybersecurity landscape. For instance, integrating AI-driven threat detection systems with user-friendly interfaces can enhance security while promoting user compliance and awareness.

2 Literature research methodology

To synthesize, integrate, and critically evaluate existing knowledge on human factors in cybersecurity, a comprehensive literature review was conducted. Following the systematic guidelines outlined by Cooper [8], this review adopts a rigorous approach to ensure reliability, relevance, and interdisciplinary coverage of the selected studies.

2.1 Literature search process

The literature search was conducted using the University of Canberra (UC) Library's discovery platform (https://www.canberra.edu.au/library), which provided access to peer-reviewed journal articles and conference proceedings through databases such as IEEE Xplore, ProQuest, EBSCOhost, ScienceDirect, SpringerLink, SAGE Journals Online, Taylor and Francis Online, ACM Digital Library, Scopus, and Web of Science. Only studies with full-text availability were considered for inclusion. Recognizing the rapid evolution of cybersecurity-related research, we limited the search to articles published between 2011 and 2024, ensuring a focus on contemporary developments and trends.

The search string used was:

("Human Factor" AND "Cybersecurity") OR ("Human Factor" AND "Cyber Security")

This targeted studies explicitly addressing human factors within cybersecurity or information security domains. Articles published in English and subjected to peer review were prioritized to ensure quality and accessibility. Initially, we received total of 128 peer review article which is accessed through UC Library.

2.2 Inclusion and exclusion criteria

To ensure the relevance and rigor of the selected studies in this survey, we employed a systematic four-step filtering process.

The first step involved initial screening to eliminate articles that were not peer-reviewed journal or conference papers, including reports, magazines, and news articles. Research-in-progress and follow-up studies were also excluded. This step was critical to maintaining the quality of the analysis and reduced the pool of studies by 28, leaving a total of 100 articles for further evaluation.

The second step focused on reviewing the abstracts of the remaining articles. Studies without explicit mention of cybersecurity and human factors in their abstracts were excluded to ensure that only papers with substantive focus on these topics were retained. This filtering removed 8 additional papers, resulting in 92 studies.

The third step assessed the content relevance of the articles. Studies primarily centred on technical cybersecurity solutions, such as ontologies, infrastructure or organizational systems, or domain-specific cybersecurity topics (e.g., banking or maritime) without sufficient emphasis on human behaviour were excluded. Follow-up studies replicating prior research with minimal new content were also removed. This rigorous review excluded 14 more studies, refining the selection to 78.

In the final step, a quality review was conducted. Articles were manually reviewed to ensure representation across diverse disciplines and focus areas pertinent to human factors in cybersecurity. While citation counts were not used as a criterion for inclusion, the selection prioritized studies that contributed valuable insights into psychological, organizational, and socio-technical dimensions. The selected studies spanned fields such as computer science, behavioural science, psychology, sociology, criminology, education, ethics, library and information sciences, industrial studies, and life sciences. This interdisciplinary approach enriched the scope of the review. After completing this filtering process, a total of 41 studies were finalized for inclusion. These studies formed the foundation for an in-depth exploration of human factors in cybersecurity, aligned with the overarching aim of this survey to provide an interdisciplinary understanding of the subject.

2.3 Article profile cards

To ensure a structured and comprehensive analysis, each selected study was documented using a study profile card. Inspired by methodologies proposed and used by King and Torkzadeh [9] and Jeong, et al. [10], these profile cards captured critical information, including: (a) Research Objectives; (b) Research Questions; (c) Key words; (d) Research Methodology; (e) Theoretical Framework(s); (f) Limitations; (g) Discipline; (h) Focus Area; and (i) Summary of Findings like an example provided in the Table 1. This systematic approach facilitated thematic synthesis, helping identify research trends, gaps, and insights across the reviewed articles.

Table 1 Example of a article profile card

Category	Details
Paper title	Botching human factors in cybersecurity in business organizations
Authors	Calvin noble [11]
Publication	Holistica, 2018
Purpose/research questions	The purpose of this paper is to analyse the impact of human errors in cybersecurity and propose strategies to address them effectively
	Research question: Why do organizations fail to address human factors in cybersecurity?
Summary of findings	Human-enabled errors account for the majority of cybersecurity breaches. Organizations over-rely on technology and fail to incorporate human factors specialists into their cybersecurity strategies, leading to increased vulnerabilities
Keywords	Information security, cybersecurity, human factors, technological determinism, human-centered cybersecurity, human-enabled errors, technology
Research methodology	Literature review and policy analysis to evaluate the role of human factors in cybersecurity failures
Theoretical framework	Human-centered cybersecurity approaches, emphasizing the integration of behavioural insights into risk management
Discipline	Cybersecurity, management science
Limitations and gaps	Over-reliance on literature without empirical testing
	Limited focus on specific industries or contexts
Focus area of research	Employee behaviour and organizational risk management

2.4 Scope and interdisciplinary insights

The selected articles reflect a broad range of disciplines, underscoring the interdisciplinary essence of human factors research in cybersecurity. While well-represented fields such as computer science and information systems formed the foundation, the review extended into: Behavioural Science and Psychology: Examining trust, stress, cognitive overload, and personality traits in cybersecurity contexts; Sociology and Criminology: Exploring social engineering tactics, authority exploitation, and trust dynamics; Education and Training: Investigating the effectiveness of gamified learning and adaptive training programs in fostering cybersecurity awareness; Library and Information Sciences: Highlighting the role of libraries in promoting community-level digital literacy and cybersecurity practices; Industrial Studies

and Human Reliability: Focusing on human error in technical environments and its implications for system reliability; Ethics and AI: Addressing privacy concerns, ethical challenges in AI-driven cybersecurity tools, and insider threat management; Life Sciences: Considering the impact of cognitive and emotional factors, such as stress and fatigue, on cybersecurity performance. This diversity enabled a holistic understanding of human factors in cybersecurity, bridging behavioural, organizational, and technical dimensions.

2.5 Limitations of the review process

Despite its comprehensive nature, the review faced some limitations: studies published in languages other than English were excluded, potentially omitting valuable non-English research; articles from grey literature or industry reports, which could provide practical insights, were not considered; the time frame (2011–2024) might exclude foundational studies conducted before 2011 that could offer historical context.

This methodology ensures a systematic, comprehensive, and interdisciplinary approach to reviewing human factors in cybersecurity. By synthesizing insights from 41 high-quality studies, the review provides a robust foundation for understanding the complex interplay between human behaviour, organizational strategies, and technical systems in cybersecurity.

3 Analysis

This section organizes the insights from the 41 selected studies into key thematic areas, highlighting critical aspects of research and development in the field of human factors in cybersecurity. The thematic analysis based on article profile card showed selected studies explores psychological dimensions, training and education, organizational culture, social engineering and phishing, ethical and cultural considerations, and the integration of human and technical systems.

1. Psychological dimensions in cybersecurity

Psychological factors significantly influence cybersecurity decision-making, shaping both vulnerability and resilience. Key findings include:

Stress and Cognitive Fatigue: Security Operations Centre (SOC) analysts often experience high cognitive load, increasing the likelihood of errors [11].

Personality Traits & Phishing Susceptibility: Neuroticism and overconfidence correlate with increased vulnerability to phishing attacks [12].

Table 2 Psychological dimensions in cybersecurity

Study	Key findings
[11, 16]	Stress, cognitive fatigue, and human error significantly exacerbate cybersecurity vulnerabilities and contribute to SOC inefficiencies. Mitigation strategies include stress management, task allocation, and improved decision-support systems
[12, 14]	Personality traits like neuroticism and behavioural biases, such as trust and urgency, increase susceptibility to phishing and social engineering attacks. Highlighted emotional triggers, such as fear and urgency, and other psychological factors, exploited in phishing attacks
[13, 14, 18]	Trust and authority dynamics are central to social engineering campaigns. Interventions must focus on resilience to psychological manipulation tactics
[19]	Habit-formation techniques enhance passive cybersecurity behaviours (e.g., password management), while decision-support systems and automation improve performance by reducing cognitive overload
[20]	Cognitive biases impact risk perception and decision-making, increasing vulnerability to social engineering attacks
[6]	Adaptive decision-support systems alleviate cognitive strain and enhance situational awareness in high-stress environments like SOCs

Trust in Social Engineering Attacks: Attackers exploit trust dynamics to manipulate individuals, reinforcing the need for psychological resilience training [13, 14].

Mitigation Strategies: Research highlights the effectiveness of task allocation, stress management, and personalized training programs in reducing these risks [11, 15, 16].

Trust and authority dynamics, similarly, psychological manipulation techniques based on persuasion strategies are central to social engineering campaigns. Attackers frequently exploit these psychological factors to manipulate individuals, necessitating targeted interventions that build resilience and awareness [13, 14, 17]. Similarly, cognitive overload in high-demand cybersecurity environments reduces performance, highlighting the value of decision-support systems and automation in alleviating this burden [11, 15]. These findings emphasize the importance of fostering psychological resilience, understanding social engineering tactics, and tailoring interventions to address these vulnerabilities effectively (Tables 2, 3, 4, 5, 6, 7).

2. Training and education

Cybersecurity education and training programs have emerged as a critical area of focus, with studies exploring innovative approaches like gamification, adaptive learning, and

Table 3 Training and education in cybersecurity

Study	Key findings
[26, 27]	Gamification enhances engagement, knowledge retention, and scenario-based learning in cybersecurity training programs, catering to diverse demographic groups
[22, 28]	Libraries and public institutions enhance community-level cybersecurity awareness and digital literacy
[7, 29–31]	Tailored training improves adherence to best practices and addresses specific organizational needs
[16, 32]	Adaptive and discipline-specific learning tools improve personalized cybersecurity education, addressing variations in awareness across different student populations

Table 4 Organizational culture and compliance

Study	Key findings
[35]	Leadership styles emphasizing emotional intelligence foster stronger cybersecurity cultures, promoting proactive and compliant behaviours among employees
[29, 30, 33]	Mismatches between organizational policies and employee behaviours hinder compliance. Aligning policies with user capabilities and expectations is critical for mitigating risks
[31, 35, 36]	Simplified cybersecurity processes, gamification, and cultural adjustments (e.g., fostering organizational trust) reduce cyber fatigue and insider threats while encouraging compliance
[31, 35, 36]	Simplified cybersecurity processes and gamification reduce cyber fatigue, encouraging employee compliance and engagement with cybersecurity policies
[7, 38]	A strong cybersecurity culture requires integration with organizational strategies, promoting trust, transparency, and proactive cybersecurity behaviours among employees

Table 5 Social engineering and phishing

Study	Key findings
[13, 14]	Social engineering campaigns exploit trust and urgency to manipulate victims, requiring targeted interventions
[39]	Trust dynamics post-breach require rebuilding mechanisms to restore organizational resilience
[30]	Training programs focusing on cognitive awareness reduce susceptibility to manipulation and enhance resilience to social engineering tactics
[40]	Gamified training programs significantly improve resistance to spear-phishing by engaging employees in interactive, real-world scenarios
[41]	Sector-specific vulnerabilities in maritime environments highlight the importance of tailored interventions for phishing and social engineering resilience

Table 6 Ethical and cultural considerations

Study	Key findings
[42]	Ethical challenges include bias and privacy concerns in AI-driven cybersecurity tools
[24]	Culturally tailored awareness programs improve inclusivity and engagement globally
[36]	Cybersecurity frameworks integrating cultural diversity considerations enhance organizational trust, inclusivity, and resilience
[33]	Addressing ethical concerns in organizational policies promotes better compliance and mitigates conflicts arising from cultural and demographic differences
[22]	Libraries and public institutions play a vital role in fostering culturally sensitive cybersecurity awareness programs, particularly in underserved communities

targeted awareness campaigns. Gamification has been shown to enhance engagement and retention in training programs, particularly among diverse demographic groups [16, 21]. Libraries and public institutions have also been identified as pivotal in promoting digital literacy and community-level cybersecurity awareness, especially in underserved populations [22, 23].

Tailored training programs that align with user behaviours and organizational needs are essential for improving cybersecurity awareness and adherence to best practices [24, 25]. Despite these advancements, challenges remain in addressing generational differences, diverse learning preferences, and the long-term retention of training outcomes. Future initiatives should aim to develop adaptive frameworks that accommodate these variations.

3. Organizational culture and compliance

Organizational culture plays a pivotal role in shaping cybersecurity practices and compliance. Research highlights leadership styles, policy alignment, and cultural integration as key factors influencing employee behaviour and adherence to cybersecurity protocols [33, 34]. Leadership styles that emphasize emotional intelligence foster a positive cybersecurity culture, encouraging proactive and compliant behaviours among employees. Leaders who demonstrate empathy, effective communication, and inclusivity inspire trust and enhance the overall organizational commitment to cybersecurity practices [35].

However, mismatches between organizational policies and employee behaviours often act as significant barriers to compliance. Studies emphasize the need to align cybersecu-

Table 7 Integration of human and technical systems

Study	Key findings
[44, 45]	Human error is a significant factor in system failures, requiring improved alignment of human and technical systems
[41]	Socio-technical frameworks enhance resilience by bridging behavioural and technical gaps
[46]	Machine learning models tailored to user behaviour enhance technical system adaptability, reducing error rates and improving user compliance
[47]	AI-based frameworks improve real-time threat detection and SOC operations, emphasizing the importance of human oversight in automated cybersecurity systems
[7, 36, 48]	Integrated human and technical approaches address insider threats by aligning cultural adjustments with technical solutions to balance privacy and security
[6]	Adaptive decision-support systems reduce cognitive strain in high-stress environments, enabling better alignment between technical defences and human behaviours

rity policies with employees' capabilities and expectations to mitigate risks and reduce non-compliance. Policies that fail to consider employee behaviour or workplace realities often lead to violations and poor adherence [33, 34].

Addressing insider threats requires a multifaceted approach that integrates technical detection systems with cultural and behavioural adjustments. Organizations that foster trust, transparency, and engagement are better equipped to prevent insider threats while maintaining a balance between privacy and security [36].

Simplified processes and gamification have also been identified as effective strategies to combat cybersecurity fatigue and increase compliance. By reducing complexity in protocols and introducing engaging elements, employees are more likely to adopt secure practices and follow organizational guidelines [37]. These insights highlight the importance of an interdisciplinary approach that combines organizational behaviour, psychology, and technical measures to create a cybersecurity culture that supports both security and compliance. Organizations must prioritize leadership development, user-centric policies, and innovative compliance mechanisms to address the evolving challenges in cybersecurity.

4. Social engineering and phishing

Social engineering tactics, particularly phishing, are a significant research focus within human factors in cybersecurity. These campaigns exploit psychological factors such as urgency, trust, and authority, requiring targeted interventions to reduce susceptibility [17]. Attackers frequently leverage urgency and authority to manipulate victims into divulging sensitive information, necessitating awareness programs addressing these vulnerabilities [13, 14]. Interpersonal and organizational trust were identified as recurring factors enabling social engineering attacks, underscoring the need for trust-rebuilding mechanisms post-breach [12, 13, 39].

5. Ethical and cultural considerations

Ethical challenges and cultural diversity are increasingly recognized as integral to cybersecurity practices, particularly in the context of AI and global awareness programs. Privacy concerns, transparency, and bias in AI-driven cybersecurity tools are persistent issues highlighted in recent studies [42, 43]. Tailored cybersecurity awareness programs that account for cultural diversity have been found to improve inclusivity and relevance [24, 36]. Addressing these considerations will require standardized frameworks to measure ethical compliance and cultural effectiveness in cybersecurity strategies.

6. Integration of human and technical systems

The interplay between human factors and technical systems remains a critical research area, emphasizing the need for socio-technical frameworks. Human error has been identified as a major contributor to technical system failures, particularly in high-risk environments like industrial settings [44, 45]. Research also highlights the value of socio-technical systems that align human behaviour with adaptive technical defences, bridging the gap between organizational processes and technical solutions [23, 41].

This thematic organization captures the multifaceted dimensions of human factors in cybersecurity. By categorizing the findings into psychological, educational, organizational, social engineering, ethical, and socio-technical themes, it provides a comprehensive foundation for understanding the current state of research. These insights inform the gaps and future directions discussed in subsequent sections, emphasizing the need for interdisciplinary approaches to enhance cybersecurity resilience.

4 Theoretical background

The role of human factors in cybersecurity has gained significant attention as technological advancements alone have proven insufficient to mitigate vulnerabilities stemming from human behaviour. Theoretical frameworks provide the foundation for understanding how human and organizational dimensions interact with technical systems, shaping cybersecurity resilience. This section explores key theories and models relevant to our literature review theme and derived from interdisciplinary research to contextualize human-centric cybersecurity strategies.

1. Protection motivation theory (PMT)

PMT, first introduced by Rogers [49] in the health sciences, has been extensively adapted in cybersecurity research to understand protective behaviours. This theory posits that individuals assess threats based on perceived severity, vulnerability, response efficacy, and self-efficacy. Studies such as those by Sulaiman et al. [33] and Siddiqi et al. [14] demonstrate the relevance of PMT in predicting user compliance with cybersecurity policies. Extensions of PMT have incorporated variables like rewards, facilitating conditions, and psychological capacities to enhance its applicability in diverse cybersecurity contexts.

2. Theory of planned behaviour (TPB)

TPB explains behavioural intentions through attitudes, subjective norms, and perceived behavioural control. This theory has been integrated with PMT to provide a comprehensive understanding of cybersecurity behaviours, as highlighted by Sulaiman et al. [33]. TPB emphasizes the social and psychological factors influencing user behaviour, making it particularly useful in addressing compliance and violation dynamics within organizations.

3. General deterrence theory (GDT)

GDT suggests that the certainty, severity, and swiftness of sanctions deter deviant behaviour. In cybersecurity, this theory is applied to design policies that discourage violations through effective deterrence mechanisms. Studies by Georgiadou et al. [36] and Sulaiman et al. [33] have demonstrated the integration of GDT with other models, such as Neutralization Theory, to address insider threats and policy violations.

4. Cybersecurity culture framework

The Cybersecurity Culture Framework, proposed by Georgiadou Georgiadou et al. [36], highlights the interplay of individual attitudes, organizational policies, and cultural factors in shaping security behaviours. This framework stresses the need for a holistic approach that integrates technical defences with human and organizational dimensions. It provides a basis for assessing readiness and fostering a proactive security culture.

5. Cognitive and psychological theories

Cognitive theories, such as Singh and Cheema [20] "Cognitive Shield" model, focus on enhancing decision-making and mental preparedness to counteract cyber threats. Psychological constructs, including trust, cognitive biases, and emotional resilience, have been identified as critical factors influencing user behaviour. Nobles [48] and Siddiqi et al. [14] have highlighted the exploitation of psychological vulnerabilities in social engineering attacks, emphasizing the need for targeted training and awareness programs.

6. Socio-technical integration

The interplay between human and technical systems is central to cybersecurity resilience. Socio-technical frameworks align user behaviour with adaptive technical defences, addressing vulnerabilities caused by human error. Research by Nobles [48] and Georgiadou et al. [36] advocates for integrating human reliability models and decision-support systems to mitigate risks and enhance organizational performance.

7. Ethical and cultural considerations

Ethical frameworks address challenges in AI-driven cybersecurity systems, such as bias, transparency, and user privacy. Nguyen [42] and Georgiadou et al. [7] emphasize culturally tailored awareness programs and ethical compliance measures to enhance inclusivity and trust in cybersecurity practices.

8. Organizational behaviour and compliance

Leadership, policy alignment, and organizational culture significantly influence cybersecurity compliance. Research by Sulaiman et al. [33] highlights the importance of aligning policies with employee capabilities and expectations. Simplifying processes and fostering trust within organizations are critical strategies for reducing cybersecurity fatigue and improving adherence to security protocols.

9. Social engineering and human vulnerabilities

Social engineering exploits psychological factors such as trust, urgency, persuasion techniques and authority. Studies by Siddiqi et al. [14] and Guitton [13] emphasize the importance of resilience-building interventions to counter manipulation tactics. Tailored training programs addressing these vulnerabilities are essential for enhancing organizational security.

By integrating psychological, organizational, and socio-technical dimensions, these theories contribute to developing human-centric strategies that enhance resilience and mitigate risks. This interdisciplinary approach highlights the need for continuous research to adapt frameworks to evolving threats and diverse contexts.

5 Research gaps and future directions

The comprehensive review of 41 studies has revealed critical gaps in the research on human factors in cybersecurity, highlighting the need for deeper exploration and a more integrated approach to address the interplay of human behaviour, organizational culture, and technical systems. This section elaborates on the key research gaps and outlines actionable directions for future studies, offering insights into how these challenges can be addressed.

5.1 Research gaps

1. Lack of Longitudinal Studies

A significant gap in the literature is the focus on short-term interventions and immediate outcomes, with little attention paid to long-term impacts and sustainability. For instance, studies on gamified training programs [21] and stress management among SOC analysts [48] provide valuable short-term insights but fail to assess their enduring effectiveness. This lack of follow-up data hinders the ability to determine whether such interventions lead to sustained behavioural changes or improved cybersecurity resilience over time.

2. Underrepresentation of Underserved Communities Research in cybersecurity has primarily targeted urban and well-connected populations, often neglecting rural, developing countries and underserved communities. While studies on libraries and community-level digital literacy [23, 50] suggest promising outreach strategies, they fail to address the unique challenges faced by marginalized groups, such as limited access to technology, resources, and training.

3. Limited Exploration of Non-Digital Social Engineering The focus on phishing and online scams overshadows the significant risks posed by non-digital manipulation techniques, including physical social engineering and insider threats. Studies like [13, 14] highlight the prevalence of trust and authority exploitation but lack comprehensive frameworks to address non-digital vulnerabilities. This gap leaves organizations unprepared for manipulation tactics that exploit interpersonal dynamics in physical settings.

4. Ethical and Privacy Concerns in AI Integration While AI-driven cybersecurity tools have revolutionized threat detection and response, ethical issues surrounding transparency, bias, and user privacy remain unresolved. Studies [42, 43] point to the potential for AI systems to reinforce biases and erode trust, yet actionable frameworks to address these concerns are still underdeveloped.

5. Narrow Focus on Psychological Constructs Research has largely concentrated on specific traits such as neuroticism and trust [12] in relation to phishing susceptibility. Broader psychological dimensions, including emotional resilience, adaptability, and recovery from cybersecurity incidents, are rarely explored. This narrow focus limits the ability to design interventions that address the full spectrum of psychological vulnerabilities. ***6. Gaps in Defining and Measuring Cybersecurity Culture*** Despite its recognized importance, cybersecurity culture lacks a universally accepted definition, clear metrics, and scalable evaluation frameworks. Studies like [33, 34] emphasize the role of culture in fostering cybersecurity resilience but do not provide concrete tools for its assessment and implementation across diverse industries.

7. Fragmentation in Socio-Technical Research Human and technical factors are often studied in isolation, overlooking their interdependence. For example, human reliability studies [23, 44] highlight error patterns but fail to consider how technical systems could adapt to mitigate these errors. This fragmentation limits the development of holistic frameworks that integrate socio-technical insights.

8. Underexplored Generational and Cultural Diversity Research has inadequately addressed how generational differences, and cultural factors shape cybersecurity practices and vulnerabilities. Programs tailored to specific demographics, such as those proposed by Park et al. [24], remain scarce and unvalidated, leaving critical gaps in inclusivity and relevance.

5.2 Future directions

1. Longitudinal Studies on Human Factors Future studies must prioritize longitudinal research to evaluate the sustained effectiveness of cybersecurity interventions, such as stress management, gamified training, and personalized education programs. This will provide insights into long-term behavioural changes and their impact on organizational resilience.

2. Broadening Outreach to Underserved Communities Efforts should be directed toward developing cybersecurity awareness initiatives for underserved populations. Leveraging local institutions like libraries and collaborating with NGOs can help bridge the digital divide, as suggested by Kont [22], Kont [50] and Bella [23].

3. Frameworks for Ethical AI Integration Actionable frameworks are needed to address ethical challenges in AI-driven cybersecurity systems. These frameworks should focus on mitigating bias, ensuring transparency, and safeguarding user privacy, as highlighted by Nguyen [42]. 4. Expanding Psychological Research Exploring broader psychological constructs, such as emotional intelligence, resilience, and adaptability, is essential for designing effective cybersecurity interventions. Tailored programs should address these factors to enhance user compliance and decision-making [12, 48].

5. Developing Cybersecurity Culture Metrics Standardized metrics for evaluating and promoting cybersecurity culture across industries should be developed. These metrics

must incorporate insights from organizational behaviour, employee engagement, and cultural diversity [34].

6. Advancing Socio-Technical Integration Future research should empirically test socio-technical frameworks that align human behaviour with adaptive technical systems. Research should be focused on decision-support systems which harmonize with human decision-making with technical defences, while Ryczyński [44] highlights the importance of integrating human reliability models to address cognitive fatigue and reduce error rates.

7. Catering to Generational and Cultural Diversity Cybersecurity training programs must be tailored to address generational differences and cultural nuances. Leveraging data from studies like Parkdiverse user groups.

8. Investigating Non-Digital Social Engineering In-depth studies on physical social engineering techniques and insider threats are needed to develop comprehensive prevention strategies. Siddiqi et al. [14] and Guitton [13] provide valuable starting points for this research.

9. Cross-Sectoral Validation of Cybersecurity Frameworks Extending and testing cybersecurity models across industries, such as maritime [41], banking [51], healthcare, and education, will ensure their adaptability and scalability.

10. Stress and Cognitive Load Management Interventions aimed at reducing stress and cognitive overload among SOC analysts and other cybersecurity professionals must be developed and validated to improve performance in high-pressure scenarios [11].

Addressing these research gaps will require interdisciplinary collaboration that integrates insights from psychology, sociology, education, and technology. By pursuing the proposed directions, researchers and practitioners can advance the field of human factors in cybersecurity, fostering a more inclusive, adaptable, and resilient digital ecosystem. Aligning future studies with real-world applicability will be crucial in achieving meaningful improvements in cybersecurity practices globally. This roadmap provides a foundation for continued innovation and effective integration of human factors into cybersecurity frameworks.

6 Proposed framework

Based on the systematic review and thematic analysis of the 41 different studies, identified research gaps, and future directions, this paper proposes an interdisciplinary Human-Centric Cybersecurity Framework to address the complex interplay of human factors, technical systems, and organizational strategies. This framework synthesizes insights from psychology, sociology, education, organizational behaviour, and technology, creating a holistic approach to enhance cybersecurity resilience. The framework emphasizes adaptability, inclusivity, and practicality, ensuring its relevance across diverse environments.

6.1 Core components of the framework

The proposed framework is built around four interconnected pillars, each targeting a critical dimension of human factors in cybersecurity:

1. Psychological Dimensions

This pillar focuses on addressing individual psychological factors that influence cybersecurity behaviours and decision-making.

Behavioural Insights: Interventions must be tailored to user-specific traits such as trust perception, cognitive adaptability, and personality characteristics. For example, studies show that traits like neuroticism and overconfidence can increase phishing susceptibility [12, 48].

Stress and Fatigue Management: Tools and workflows designed to reduce decision fatigue and stress are essential, particularly for high-stakes environments like SOCs [44].

Emotional Intelligence Training: Programs should equip leaders and employees with emotional intelligence skills to foster team resilience, enhance collaboration, and build cybersecurity awareness [35].

2. Training and Awareness

This pillar emphasizes the importance of education and training to enhance cybersecurity literacy and preparedness across diverse populations.

Gamified Learning: Gamification strategies can improve user engagement and retention, as shown in diverse demographic groups [16, 21].

Digital Literacy in Communities: Institutions like libraries can bridge digital divides by promoting cybersecurity awareness, particularly in underserved and rural populations [23, 50]. Kont [50] focuses on libraries in Estonia as hubs for digital literacy and cybersecurity education, while Kont [28] highlights the broader role of libraries in addressing cybersecurity gaps in rural areas and disadvantaged communities.

Generational and Cultural Tailoring: Training programs must consider demographic diversity, integrating cultural adaptation and generational differences for maximum inclusivity [24].

3. Organizational Culture and Compliance

This pillar aims to align organizational policies, leadership, and culture with human-centric cybersecurity practices.

Policy-Behaviour Alignment: Cybersecurity policies must resonate with employee behaviours and capabilities to improve compliance and reduce risk [34].

Insider Threat Mitigation: Effective mitigation strategies should integrate technical measures with cultural and behavioural interventions [31].

Leadership Development: Leadership styles that prioritize emotional intelligence and foster a proactive cybersecurity

culture can significantly improve compliance and organizational resilience [35].

4. Socio-Technical Integration

This pillar focuses on harmonizing human behaviour with technical systems to mitigate risks and optimize system performance.

Adaptive Decision-Support Systems: Tools designed to reduce cognitive overload and align with human decision-making processes can improve performance in high-stress scenarios. Jeong [10] highlights the importance of integrating socio-technical insights into these systems, ensuring that human behavioural tendencies and limitations are considered in their design.

Human Reliability Models: Systems that account for human error, particularly in high-risk environments like SOCs, are critical for reducing vulnerabilities [44].

AI Ethics and Privacy: Addressing ethical challenges in AI-driven cybersecurity tools, such as transparency, bias, and user-centric design, is essential for building trust.

Non-Digital Social Engineering Prevention

Physical Vulnerability Mitigation: Incorporating frameworks addressing physical social engineering tactics, such as access control training and organizational policy improvements.

Human Interaction Training: Equipping employees with skills to identify and respond to manipulation tactics in interpersonal settings. Guitton [13] emphasizes the need for proactive training to combat manipulation, while Siddiqi et al. [14] highlights the psychological underpinnings exploited in interpersonal social engineering.

Inclusivity and Cultural Adaptation

Generational and Cultural Tailoring: Training programs should address generational differences and cultural variations to promote inclusivity and effectiveness.

Global Adaptability: Testing and adapting cybersecurity practices across diverse cultural and organizational settings ensures relevance and effectiveness.

Longitudinal and Cross-Sectoral Evaluation

Framework Validation: Empirical testing of cybersecurity frameworks across diverse industries such as healthcare, maritime, and finance is critical for assessing scalability and adaptability.

Long-Term Impact Studies: Longitudinal studies are essential for measuring sustained behavioural changes and organizational resilience. Linkov [50] discuss their importance in fostering long-term adherence to cybersecurity practices, while Creazza et al. [52] emphasizes the role of such studies in evaluating sector-specific challenges.

6.2 Framework implementation

The framework is designed to be iterative and dynamic, implemented through the following steps:

1. Assessment Phase

- Conduct behavioural audits to identify psychological, cultural, and organizational vulnerabilities.
- Map existing socio-technical systems to pinpoint misalignments between human behaviours and technical infrastructure.

2. Design Phase

- Develop tailored training modules leveraging gamification, cultural adaptation, and scenario-based learning.
- Create decision-support systems that align with users' cognitive and emotional needs to improve cybersecurity decision-making.
- Integrate human reliability and ethical AI considerations into technical system designs.

3. Implementation Phase

- Roll out training programs across diverse demographic groups, prioritizing inclusivity and accessibility.
- Deploy socio-technical systems with iterative feedback loops to adapt to real-time challenges.

4. Evaluation Phase

- Use both cross-sectional and longitudinal studies to evaluate the effectiveness and sustainability of interventions, measuring changes in behaviour, system performance, and user engagement over time.
- Adjust framework components based on empirical findings and emerging cybersecurity challenges.

6.3 Expected outcomes

This framework offers several tangible benefits:

- Enhanced Cybersecurity Literacy: Improved awareness and compliance across all user groups, including underserved and rural communities.
- Resilient Workforces: Reduced stress and decision fatigue among SOC analysts and employees.
- Culturally Inclusive Practices: Tailored programs that respect cultural and generational diversity, fostering greater engagement and inclusivity.
- Integrated Systems: Socio-technical systems that align technical defences with human behaviours, reducing errors and vulnerabilities.
- Ethically Sound AI Applications: Transparent and user-centric AI tools that address privacy and ethical concerns.

The proposed Human-Centric Cybersecurity Framework integrates psychological, educational, organizational, and technical dimensions to create a cohesive approach to enhancing cybersecurity resilience. By addressing key human factors such as behaviour, stress, and cultural diversity, this framework provides actionable strategies for mitigating risks and improving the effectiveness of cybersecurity practices across diverse environments. It serves as a roadmap for researchers and practitioners to build adaptive, inclusive, and resilient cybersecurity ecosystems.

7 Limitation

Despite the contributions of this study, several limitations should be acknowledged. First, the proposed framework is based on literature synthesis and has not yet undergone empirical validation in real-world settings. Future research should conduct experimental studies and longitudinal assessments to measure its practical effectiveness. Second, while the study highlights the role of cybersecurity culture, standardized metrics for evaluating and implementing it across different organizational contexts remain underdeveloped. Third, the focus on digital threats, such as phishing and AI-driven attacks, leaves non-digital social engineering tactics, including insider threats and physical breaches, underexplored. Additionally, emerging technologies like blockchain, quantum computing, and IoT security require further integration into human-centric cybersecurity frameworks. Addressing these limitations will enhance the applicability and effectiveness of human-centric cybersecurity strategies.

8 Conclusion

This paper has provided a literature survey based synthesis of research on human factors in cybersecurity, emphasizing the critical role that behavioural, psychological, and organizational dimensions play in securing digital environments. The increasing complexity of cyber threats requires an approach that goes beyond purely technological solutions, recognizing that vulnerabilities stemming from human behaviour often undermine even the most advanced technical defences.

Through a detailed analysis of 41 selected studies, we identified several key themes, including the impact of psychological factors such as stress, cognitive overload, persuasion strategies, and trust on cybersecurity, the importance of tailored training and awareness programs, and the vital role of organizational culture and leadership in shaping compliance and resilience. These insights highlight the need for interdisciplinary approaches that integrate human factors into the design and implementation of cybersecurity strategies. However, despite significant progress, gaps remain, particularly in understanding long-term behavioural changes, addressing the needs of underserved communities, integrating socio-technical approaches, and managing the ethical challenges posed by AI-driven security systems.

To address these gaps, this paper proposes a Human-Centric Cybersecurity Framework that integrates psychological, educational, organizational, and technical dimensions. This framework offers actionable strategies for enhancing cybersecurity resilience by focusing on habit formation, stress management, gamified training, and the alignment of socio-technical systems. By adopting this approach, organizations can better mitigate human-driven vulnerabilities, creating security systems that are not only technologically robust but also aligned with the complexities of human behaviour.

Looking forward, it is clear that the future of cybersecurity will increasingly depend on the integration of human factors into security practices. Researchers and practitioners must work collaboratively to develop systems that address both the technical and human aspects of cybersecurity. For instance, organizations must prioritize the development of inclusive training programs that account for generational and cultural differences, ensuring that all users are equipped with the knowledge and tools to protect themselves in a rapidly evolving threat landscape.

This study calls for a shift from viewing users as the weakest link to recognizing them as an integral part of the cybersecurity defence. By leveraging interdisciplinary strategies and focusing on human-centric approaches, we can create more resilient, adaptive, and inclusive cybersecurity ecosystems. This will not only enhance the security of digital systems but also foster a culture of trust and collaboration, empowering users to play an active role in safeguarding the digital world.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Kadena, E., Gupi, M.: Human factors in cybersecurity. Secur. Sci. J. **2**, 51–64 (2021)
2. Berman, J., et al.: Twitter Investigation Report (2020). https://www.dfs.ny.gov/Twitter_Report. Accessed 13 Dec 2024
3. Corradini, I.: Security: Human Nature and Behaviour (2020)
4. Rohan, R., Funilkul, S., Pal, D., Chutimaskul, W.: Understanding of human factors in cybersecurity: a systematic literature review. In: 2021 International Conference on Computational Performance Evaluation (ComPE), pp. 133–140 (2021)
5. Robinson, N.: Human factors security engineering: the future of cybersecurity teams. EDPACS **67**, 1–17 (2023)
6. Pollini, A., et al.: Leveraging human factors in cybersecurity: an integrated methodological approach. Cognit. Technol. Work **24**(2), 371–390 (2021). https://doi.org/10.1007/s10111-021-00683-y
7. Georgiadou, A., Mouzakitis, S., Bounas, K., Askounis, D.: A cybersecurity culture framework for assessing organization readiness. J. Comput. Inf. Syst. **62**(3), 452–462 (2022). https://doi.org/10.1080/08874417.2020.1845583
8. Cooper, H.: Synthesizing Research: A Guide for Literature Reviews. Sage, Newcastle upon Tyne (1998)
9. King, W.R., Torkzadeh, G.: Information systems offshoring: research status and issues. MIS Q. **32**(2), 205–225 (2008)
10. Jeong, J., Mihelcic, J., Oliver, G., Rudolph, C.: Towards an improved understanding of human factors in cybersecurity. In: 2019 IEEE 5th International Conference on Collaboration and Internet Computing (CIC), pp. 338–345. IEEE (2019)
11. Chamkar, S.A., Maleh, Y., Gherabi, N.: The human factor capabilities in security operation Center (SOC). EDPACS **66**(1), 1–14 (2022). https://doi.org/10.1080/07366981.2021.1977026
12. López-Aguilar, P., Solanas, A.: The role of phishing victims' neuroticism: reasons behind the lack of consensus. Int. J. Inf. Secur. Cybercrime (Online) **10**(2), 75–80 (2021). https://doi.org/10.19107/IJISC.2021.02.07
13. Guitton, M.J.: Manipulation through online sexual behavior: exemplifying the importance of human factor in intelligence and counterintelligence in the big data era. Int. J. Intell. Secur. Public Aff. **21**(2), 117–142 (2019). https://doi.org/10.1080/23800992.2019.1649122
14. Siddiqi, M.A., Pak, W., Siddiqi, M.A.: A study on the psychology of social engineering-based cyberattacks and existing countermeasures. Appl. Sci. **12**(12), 6042 (2022). https://doi.org/10.3390/app12126042
15. Ani, U.D., He, H., Tiwari, A.: Human factor security: evaluating the cybersecurity capacity of the industrial workforce. J. Syst. Inf. Technol. **21**(1), 2–35 (2019). https://doi.org/10.1108/JSIT-02-2018-0028
16. Tan, Z., Beuran, R., Hasegawa, S., Jiang, W., Zhao, M., Tan, Y.: Adaptive security awareness training using linked open data datasets. Educ. Inf. Technol. **25**(6), 5235–5259 (2020). https://doi.org/10.1007/s10639-020-10155-x
17. Khadka, K., Ullah, A.B., Ma, W., Marroquin, E.M., Alem, Y.: A survey on the principles of persuasion as a social engineering strategy in phishing. In: 2023 IEEE 22nd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), pp. 1631–1638. IEEE (2023)
18. Greavu-Serban, V., Constantin, F.: Aspects of social engineering and its modalities to countercharge and prevent it. Inform. Econ. **26**(3/2022), 5–15 (2022). https://doi.org/10.24818/issn14531305/26.3.2022.01
19. Prümmer, J.: The role of cognition in developing successful cybersecurity training programs: passive vs. active engagement. Lecture Notes in Computer Science (2024)
20. Singh, B., Cheema, S.S.: Psychological defenses in cyberspace: unveiling the significance of cognitive shields in cybersecurity. Int. J. Adv. Netw. Appl. **16**(1), 6291–6295 (2024). https://doi.org/10.35444/IJANA.2024.16108
21. Antunes, M., Silva, C., Marques, F.: An integrated cybernetic awareness strategy to assess cybersecurity attitudes and behaviours in school context. Appl. Sci. **11**(23), 11269 (2021). https://doi.org/10.3390/app112311269
22. Kont, K.R.: Information security awareness of librarians in the Baltic countries: a comparative analysis. Baltic J. Mod. Comput. **11**(3), 450–474 (2023). https://doi.org/10.22364/bjmc.2023.11.3.07
23. Bella, G.: Out to explore the cybersecurity planet. J. Intellect. Cap. **21**(2), 291–307 (2020). https://doi.org/10.1108/JIC-05-2019-0127
24. Park, S., Lim, J., Kim, D.: The human factor of cybersecurity and the prevention and counter measures against cybercrime in South Korea. Webology **19**(2), 7962–7976 (2022)
25. Luan, K., Halvorsrud, R., Boletsis, C.: Evaluation of a tool to increase cybersecurity awareness among non-experts (SME employees). In: ICISSP 2023—9th International Conference on Information Systems Security and Privacy. SciTePress (2023)
26. Præstiin Jepsen, L., Håkon Meland, P., Kaloudi, N.: Increasing maritime cybersecurity awareness through game-based learning. J. Phys.: Conf. Ser. **2867**(1), 12050 (2024). https://doi.org/10.1088/1742-6596/2867/1/012050
27. Löffler, E., Schneider, B., Asprion, P.M., Zanwar, T.: CySecEscape 2.0—a virtual escape room to raise cybersecurity awareness. Int. J. Serious Games **8**(1), 59–70 (2021). https://doi.org/10.17083/ijsg.v8i1.413
28. Kont, K.R.: The role of libraries in creating a safer cyberspace: awareness of Estonian library employees in information security. Int. Inf. Libr. Rev. **56**(2), 135–149 (2024). https://doi.org/10.1080/10572317.2024.2315630
29. Dornheim, P., Zarnekow, R.: Determining cybersecurity culture maturity and deriving verifiable improvement measures. Inf. Comput. Secur. **32**(2), 179–196 (2024). https://doi.org/10.1108/ICS-07-2023-0116
30. Aschwanden, R., Messner, C., Höchli, B., Holenweger, G.: Employee behavior: the psychological gateway for cyberattacks. Organ. Cybersecur. J.: Pract. Process People **4**(1), 32–50 (2024). https://doi.org/10.1108/OCJ-02-2023-0004
31. Abawajy, J.: User preference of cyber security awareness delivery methods. Behav. Inf. Technol. **33**(3), 237–248 (2014). https://doi.org/10.1080/0144929X.2012.708787
32. Huraj, L., Lengyelfalusy, T., Hurajová, A., Lajčin, D.: Measuring cyber security awareness: a comparison between computer science and media science students. TEM J. **12**(2), 623–633 (2023). https://doi.org/10.18421/TEM122-05
33. Sulaiman, N.S., Fauzi, M.A., Wider, W., Rajadurai, J., Hussain, S., Harun, S.A.: Cyber-information security compliance and violation behaviour in organisations: a systematic review. Soc. Sci. (Basel) **11**(9), 386 (2022). https://doi.org/10.3390/socsci11090386
34. Polishchuk, S., Ivanchenko, E., Bernas, M., Jancarczyk, D.: Concept of cybernetic ability estimation of a human-operator for next generation aircraft "fly-by-wireless". Procedia Comput. Sci. **207**, 118–127 (2022). https://doi.org/10.1016/j.procs.2022.09.044
35. Burton, S.L., Burrell, D.N., Nobles, C., Jones, L.A.: Exploring the nexus of cybersecurity leadership, human factors, emotional intelligence, innovative work behavior, and critical leadership traits. Sci. Bull. **28**(2), 162–175 (2023). https://doi.org/10.2478/bsaft-2023-0016
36. Georgiadou, A., Mouzakitis, S., Askounis, D.: Detecting insider threat via a cyber-security culture framework. J. Comput. Inf. Syst. **62**(4), 706–716 (2022). https://doi.org/10.1080/08874417.2021.1903367

37. Lugo, R.G., Sütterlin, S., Knox, B.J., Bukauskas, L., Brilingaitė, A., Maennel, O.M.: Editorial: The human factor in cyber security education. Front. Educ. (Lausanne) (2023). https://doi.org/10.3389/feduc.2023.1277282
38. Pavlova, E.: Enhancing the organisational culture related to cyber security during the university digital transformation. Inf. Secur. **46**(3), 239–249 (2020). https://doi.org/10.11610/isij.4617
39. Searle, R., Renaud, K.: Trust and vulnerability in the cybersecurity context. In: Hawaii International Conference on System Science 2023 (2023)
40. Yasin, A., Fatima, R., JiangBin, Z., Afzal, W., Raza, S.: Can serious gaming tactics bolster spear-phishing and phishing resilience? Securing the human hacking in Information Security. Inf. Softw. Technol. **170**, 107426 (2024). https://doi.org/10.1016/j.infsof.2024.107426
41. Kanwal, K., Shi, W., Kontovas, C., Yang, Z., Chang, C.-H.: Maritime cybersecurity: Are onboard systems ready? Marit. Policy Manag. **51**(3), 484–502 (2024). https://doi.org/10.1080/03088839.2022.2124464
42. Nguyen, T.N.: Ollabench: Evaluating LLMs' Reasoning for Human-Centric Interdependent Cybersecurity (2024). https://doi.org/10.48550/arxiv.2406.06863
43. Linkov, V., Zámecník, P., Havlícková, D., Pai, C.W.: Human factors in the cybersecurity of autonomous vehicles: trends in current research. Front. Psychol. **10**(MAY), 995–995 (2019). https://doi.org/10.3389/fpsyg.2019.00995
44. Ryczyński, J.: Human factor as a determinant of reliability and safety of technical systems. J. KONBiN **49**(3), 195–220 (2019). https://doi.org/10.2478/jok-2019-0056
45. Hareide, O.S., Jøsok, Ø., Lund, M.S., Ostnes, R., Helkala, K.: Enhancing navigator competence by demonstrating maritime cyber security. J. Navig. **71**(5), 1025–1039 (2018). https://doi.org/10.1017/S0373463318000164
46. Al-Mashhour, A., Alhogail, A.: Machine-learning-based user behavior classification for improving security awareness provision. Int. J. Adv. Comput. Sci. Appl. **14**(8), 166–178 (2023). https://doi.org/10.14569/IJACSA.2023.0140819
47. Demertzis, K., Tziritas, N., Kikiras, P., Sanchez, S.L., Iliadis, L.: The next generation cognitive security operations center: adaptive analytic lambda architecture for efficient defense against adversarial attacks. Big Data Cognit. Comput. **3**(1), 1–21 (2019). https://doi.org/10.3390/bdcc3010006
48. Nobles, C.: Botching human factors in cybersecurity in business organizations. Holist.: J. Bus. Public Adm. **9**(3), 71–88 (2018). https://doi.org/10.2478/hjbpa-2018-0024
49. Rogers, R.W.: Cognitive and Physiological Processes in Fear Appeals and Attitude Change: A Revised Theory of Protection Motivation. Guilford Press, New York (1983)
50. Kont, K.-R.: Cyber literacy skills of Estonians: activities and policies for encouraging knowledge-based cyber security attitudes. Inf. Media **96**(96), 80–94 (2023). https://doi.org/10.15388/Im.2023.96.67
51. Kloba, L., Kloba, T.: Cyber threats of the banking sector in the conditions of the war in Ukraine. Fìnansovo-kreditna dìâl?nìst?: problemi teorìì ta praktiki, 5(46):19–28 (2022) https://doi.org/10.55643/fcaptp.5.46.2022.3883
52. Creazza, A., Colicchia, C., Spiezia, S., Dallari, F.: Who cares? Supply chain managers' perceptions regarding cyber supply chain risk management in the digital transformation era. Supply Chain Manag. **27**(1), 30–53 (2022). https://doi.org/10.1108/SCM-02-2020-0073

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.